Highest Common Factor of three numbers

We can apply Euclid's Division Algorithm twice to find the Highest Common Factor (HCF) of three positive integers using the following procedure.

Let $a,\ b,\ c$ be the given positive integers.

(i) Find HCF of a, b. Call it as d

$$d = (a, b)$$

(ii) Find HCF of d and c.

This will be the HCF of the three given numbers $a,\ b,\ c$

Example 2.6 Find the HCF of 396, 504, 636.

Solution To find HCF of three given numbers, first we have to find HCF of the first two numbers.

To find HCF of 396 and 504

Using Euclid's division algorithm we get $504 = 396 \times 1 + 108$

The remainder is $108 \neq 0$

Again applying Euclid's division algorithm $396 = 108 \times 3 + 72$

The remainder is $72 \neq 0$,

Again applying Euclid's division algorithm $108 = 72 \times 1 + 36$

The remainder is $36 \neq 0$,

Again applying Euclid's division algorithm $72 = 36 \times 2 + 0$

Here the remainder is zero. Therefore HCF of $396,\ 504 = 36$.

To find the HCF of 636 and 36.

Using Euclid's division algorithm we get $636 = 36 \times 17 + 24$

The remainder is $24 \neq 0$

Again applying Euclid's division algorithm $36 = 24 \times 1 + 12$

The remainder is $12 \neq 0$

Again applying Euclid's division algorithm $24 = 12 \times 2 + 0$

Here the remainder is zero. Therefore HCF of $636, 36 = 12$

Therefore Highest Common Factor of 396, 504 and 636 is 12.

Two positive integers are said to be relatively prime or co prime if their Highest Common Factor is 1.

Exercise 2.1

1. Find all positive integers, when divided by 3 leaves remainder 2.
2. A man has 532 flower pots. He wants to arrange them in rows such that each row contains 21 flower pots. Find the number of completed rows and how many flower pots are left over.

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